

Hemoglobin Balance

You have 5 L of blood with a hemoglobin concentration of 150 g/L and a hematocrit of 45% (A norma). You lose 1.5 L of blood from a leg injury, but the body quickly restores the total blood volume to 5 L by adding plasma.

Hint: Total hemoglobin is the hemoglobin remaining in the body; hematocrit is the fraction of red blood cells in the restored blood volume; hemoglobin concentration is total hemoglobin divided by total blood volume.

Hints / breadcrumbs:

1. **Total hemoglobin initially:** multiply the initial concentration by total blood volume.
2. **Blood volume restoration:** after losing blood, the body quickly restores blood volume to normal by adding fluid (plasma) — red blood cells are not replaced immediately.
3. **New hematocrit:** fraction of red blood cells in the restored blood volume.
4. **New hemoglobin concentration:** total hemoglobin divided by restored blood volume.
5. **Total hemoglobin after loss:** only the hemoglobin that remains in the body (does not get replaced immediately).